

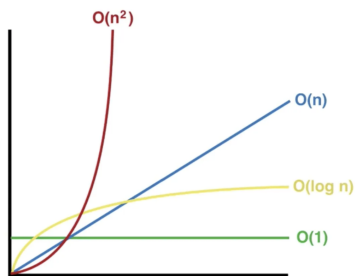
CSSE 220 – Object-Oriented Software Development
 Rose-Hulman Institute of Technology

Worksheet 17

Name (Print): _____ Section: _____

1. The efficiency of an algorithm depends on two factors: _____ complexity and _____ complexity.
2. Select the **correct** statement for **time complexity**:
 - (a) the number of times a particular instruction set is executed
 - (b) the total time taken to execute the program

3. Consider the graph below. Provide the labels



- 1) x-axis: _____ 2) y-axis: _____
4. Provide the order of the **four** time complexity families as input size increases:

5. Select the correct Big-O classification:

- (a) $3n^2$: $O(1)$ $O(n)$ $O(n^2)$
- (b) 1,000,000: $O(1)$ $O(n)$ $O(n^2)$
- (c) $2n^2 + n + 3$: $O(1)$ $O(n)$ $O(n^2)$
- (d) $\frac{n}{2} + n$: $O(1)$ $O(n)$ $O(n^2)$

6. Identify the Big-O time complexity

```

1 static int addNum(int n1,int n2){
2     int sum = n1 + n2;
3     return sum;
4 }
5 //BigO: _____
```

7. Identify the Big-O time complexity

```

1 public int sum(int[] a) {
2     int s = 0;
3     for (int x : a) {
4         s += x;
5     }
6     return s;
7 }
8 //BigO: -----

```

8. Identify the Big-O time complexity

```

1 public void twoPass(int[] a) {
2     int sum = 0;
3     for (int x : a) {
4         sum += x;
5     }
6     for (int x : a) {
7         System.out.println(x);
8     }
9 }
10 //BigO: -----

```

9. Identify the Big-O time complexity

```

1 public void printAllPairs(int[] a) {
2     for (int x : a) {
3         for (int y : a) {
4             System.out.println("(" + x + "," + y + ")");
5         }
6     }
7 }
8 //BigO: -----

```

10. Complete the table with an example for each Big-O category:

Big-O	Example
$O(1)$	
$O(\log n)$	
$O(n)$	
$O(n^2)$	

11. Before You Leave

Write one question you still have about today's topic.